

Enhancing Runtime Wi-Fi MAC Re-randomization via Temporal Jitter and Traffic Fragmentation

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Abstract—Passive MAC address observation based device tracking poses significant threats even in IEEE 802.11 wireless networks in the context of existing MAC address randomization schemes which do not guarantee continuity of identity in the course of active sessions. This work introduces and evaluates two novel techniques as part of an over-the-air runtime Wi-Fi MAC re-randomization framework. The first is called Secure-Adaptive Synchronization with Temporal Jitter (SAS-J), which makes use of cryptographic pseudorandom intervals for determining timing of synchronization, which are calculated independently at the Station and Access Points, and which make use of PTK established with the new WPA3 key management suite. This effectively nullifies potential attacks based on analysis of rotation period intervals. The second technique, referred to as Multi-Identity Traffic Fragmentation (MIRAGE-Lite), generates multiple temporary MAC addresses in each rotation cycle and distributes traffic generated by the station among multiple MAC addresses in a round-robin fashion. It further employs a sequence number masking technique to disrupt correlation attempts relying on the behavior of 802.11 sequence counters. The framework is implemented and evaluated using ns-3 for an IEEE 802.11g infrastructure-mode topology. Results show a reduction of adversary's success rate from 100% to minimal throughput loss and a mere 0.03ms overhead.

Index Terms—Wi-Fi Privacy, MAC Randomization, Traffic Analysis, IEEE 802.11, Network Security

I. INTRODUCTION

The use of wireless local area networks (WLANs) according to the IEEE 802.11 standard has become a crucial element in contemporary communication systems. Mobile devices continually send Wi-Fi frames with MAC addresses unique to each network interface. Because MAC addresses are sent in clear text in the header section of each frame and are consistent across all devices, it is possible for passive attackers listening in monitor mode to analyze WLAN traffic and record the behavior of mobile devices.

To reduce device tracking, modern operating systems implement MAC address randomization mechanisms. However, conventional randomization remains ineffective during active Wi-Fi sessions because devices typically retain the same randomized MAC address throughout the connection duration. Recently proposed over-the-air runtime Wi-Fi MAC re-randomization frameworks improve privacy by dynamically changing MAC addresses during active communication without disconnecting from the Access Point (AP).

Despite these improvements, important privacy limitations remain. The original framework performs synchronized MAC

transitions using a fixed deterministic interval T , creating predictable temporal patterns. An adversary can exploit these periodic transitions using burst interval attacks to associate disappearing and newly appearing MAC identities. Furthermore, traffic behavior characteristics such as packet timing and inter-arrival rates remain largely unchanged across cycles. To address these limitations, this work proposes two enhancements: Secure-Adaptive Synchronization with Temporal Jitter (SAS-J) and Multi-Identity Traffic Fragmentation (MIRAGE-Lite).

II. LITERATURE REVIEW

MAC address randomization has emerged as an important privacy-preserving mechanism in wireless networks. Early approaches focused primarily on periodically changing device MAC addresses during network discovery or reconnection phases to reduce long-term device tracking. However, several studies demonstrated that traditional MAC randomization fails to prevent active-session tracking because devices maintain the same randomized MAC address throughout an association period.

Fenske et al. [2] analyzed MAC randomization implementations across modern mobile devices and demonstrated multiple scenarios where tracking remained possible due to inconsistent randomization policies and protocol-level metadata leakage. Similarly, Vanhoef et al. [4] showed that Wi-Fi discovery mechanisms and management frame behaviors could bypass MAC randomization protections and enable device fingerprinting. Matte et al. [3] further demonstrated that timing analysis can defeat MAC randomization by correlating device transmission behavior across identity changes.

To address session-level privacy leakage, Jin and Papadimitratos [1] proposed an over-the-air runtime MAC re-randomization framework that dynamically changes MAC addresses during active Wi-Fi communication. Their scheme computes synchronized ephemeral MAC addresses using the WPA3 Pairwise Transient Key (PTK) and periodically updates sequence numbers and packet nonces to prevent straightforward linkage attacks. Unlike previous reconnect-based approaches, their design avoids communication disruption and maintains upper-layer transparency through MAC conversion mechanisms at both stations and access points.

Although the runtime re-randomization framework significantly improves MAC unlinkability, the authors explicitly

identify traffic analysis and fingerprinting attacks as unresolved future work. The scheme employs deterministic re-randomization intervals using a fixed period T , which introduces predictable temporal patterns observable by passive adversaries. Furthermore, the approach preserves continuous traffic behavior despite MAC transitions, allowing attackers to correlate flows using packet timing, burst structure, and inter-arrival distributions.

The work presented in this paper extends the runtime MAC re-randomization framework by addressing both deterministic leakage and behavioral flow correlation. The proposed SAS-J mechanism introduces cryptographically generated temporal jitter into synchronized MAC transitions, while MIRAGE-Lite introduces simultaneous multi-identity traffic fragmentation to disrupt traffic-flow continuity and behavioral fingerprinting.

III. METHODOLOGY

This work extends the framework proposed by Jin and Papadimitratos through two enhancements implemented within the ns-3 Wi-Fi stack.

A. Original Runtime MAC Re-randomization Logic

The baseline runtime MAC re-randomization framework dynamically changes MAC addresses during active Wi-Fi communication sessions without disconnecting the station from the Access Point. The original design generates synchronized ephemeral MAC addresses derived from the shared PTK and periodically replaces the active MAC identity during communication.

The original framework computes ephemeral MAC identities using:

$$MAC_i = H(BaseMAC \parallel PTK \parallel Epoch_i) \quad (1)$$

where $BaseMAC$ represents the original hardware MAC address, PTK is the shared Pairwise Transient Key, and $Epoch_i$ denotes the current re-randomization cycle.

Before transmission, outgoing packet headers are modified with the generated ephemeral MAC address. Upon reception, the Access Point restores the original identity before packets are delivered to upper networking layers.

Although this approach improves address unlinkability, two important limitations remain. First, MAC transitions occur at fixed deterministic intervals, enabling timing-based linkage attacks. Second, traffic-flow behavior remains continuous across re-randomization cycles, enabling behavioral fingerprinting.

B. Threat Model

The adversary considered in this work is a passive wireless observer operating in monitor mode. The attacker can capture IEEE 802.11 frames, observe MAC address transitions, analyze packet timing behavior, and perform traffic-flow correlation analysis. However, the attacker cannot decrypt WPA3 traffic or obtain the shared PTK.

C. System Design Overview

The framework operates using a shared PTK established during WPA3 authentication. Because both entities possess the same PTK, they can independently compute identical rotation schedules.

- **SAS-J:** Replaces fixed intervals with pseudo-random synchronized intervals.
- **MIRAGE-Lite:** Generates multiple ephemeral MAC identities and distributes packets via a round-robin dispatcher.

D. Secure-Adaptive Synchronization with Temporal Jitter (SAS-J)

SAS-J replaces the fixed re-randomization interval with pseudo-random synchronized intervals derived from the shared PTK. Instead of rotating MAC addresses every fixed period, both the AP and STA independently compute a pseudo-random interval during each re-randomization cycle.

The rotation scheduling logic first verifies whether the current interval has expired:

$$CurrentTime \geq NextRotationTime \quad (2)$$

If the interval expires, the cycle index is incremented:

$$CycleIndex \leftarrow CycleIndex + 1 \quad (3)$$

A synchronized pseudo-random seed is then generated using:

$$Seed_i = H(PTK \parallel CycleIndex_i) \quad (4)$$

The generated seed initializes the pseudo-random interval generator:

$$T_i = PRF(PTK \parallel CycleIndex_i) \quad (5)$$

The resulting interval is bounded as:

$$15s \leq T_i \leq 45s \quad (6)$$

The next rotation timestamp is updated as:

$$NextRotationTime = CurrentTime + T_i \quad (7)$$

Because both communicating entities possess identical PTK values and cycle indices, they independently compute the same interval without exchanging synchronization frames over the wireless medium.

The primary contribution of SAS-J is the elimination of deterministic temporal signatures. Instead of predictable transitions such as a constant interval of 30s or time T . Instead the attacker observes non-deterministic transitions such as:

$$17s \rightarrow 41s \rightarrow 22s \rightarrow 35s \quad (8)$$

This significantly weakens timing-based MAC linkage attacks.

E. MIRAGE-Lite Multi-Identity Fragmentation

Although SAS-J mitigates timing-based attacks, traffic-flow continuity may still reveal device behavior patterns. MIRAGE-Lite addresses this limitation by introducing simultaneous multi-identity communication.

Instead of maintaining a single ephemeral MAC identity during a rotation cycle, the framework generates two independent virtual MAC identities:

$$vMAC_n = H(BaseMAC \parallel PTK \parallel CycleIndex \parallel ID_n) \quad (9)$$

where ID_n distinguishes the generated virtual identities.

Outgoing packets are distributed between the generated identities using a lightweight round-robin dispatcher:

$$Packet_n \rightarrow vMAC_{(n \bmod 2)} \quad (10)$$

The dispatcher alternates packet transmission between the two identities:

$$Packet_1 \rightarrow vMAC_0 \quad (11)$$

$$Packet_2 \rightarrow vMAC_1 \quad (12)$$

$$Packet_3 \rightarrow vMAC_0 \quad (13)$$

This transforms a single continuous traffic stream into multiple fragmented lower-rate flows. The mechanism weakens packet timing correlation, throughput fingerprinting, and machine-learning-based flow linkage attacks.

F. Sequence Number Reset

IEEE 802.11 sequence numbers increase continuously during communication and may leak identity continuity across MAC transitions. To mitigate this leakage, the proposed introduces a scheme for sequence number masking.

At the beginning of each re-randomization cycle, the sequence number of the first outgoing packet is stored as an offset and the sequence number is reset to ensure that attacker cannot observe its continuity to track a device :

$$SN_{offset} = SN_{initial} \quad (14)$$

Outgoing packets use normalized sequence numbers:

$$SN_{masked} = (SN_{current} - SN_{offset}) \bmod 4096 \quad (15)$$

Upon reception, the original sequence number is restored before delivery to upper networking layers:

$$SN_{original} = (SN_{masked} + SN_{offset}) \bmod 4096 \quad (16)$$

This prevents sequence number continuity leakage while preserving protocol transparency.

G. Optimized MAC Re-randomization Algorithm

The following logic outlines the interception of the transmission and reception paths to implement SAS-J and MIRAGE-Lite.

Algorithm 1 Optimized Wi-Fi MAC Re-randomization

```

1: Function CheckAndRotate(CurrentTime):
2:   if CurrentTime  $\geq$  next_rotation_time then
3:     cycle_index  $\leftarrow$  cycle_index + 1
4:     Seed  $\leftarrow$  Hash(PTK  $\parallel$  cycle_index)
5:     RandomInterval  $\leftarrow$  GenerateRandom(Seed, 15.0, 45.0)
6:     next_rotation_time  $\leftarrow$  CurrentTime + RandomInterval
7:     vMacs[0]  $\leftarrow$  Hash(BaseMac + PTK + cycle_index + "ID_0")
8:     vMacs[1]  $\leftarrow$  Hash(BaseMac + PTK + cycle_index + "ID_1")
9:     sn_reset_needed  $\leftarrow$  True
10:   end if
11: Function ProcessOutgoingPacket(Packet):
12:   CheckAndRotate(Now())
13:   FakeMacToUse  $\leftarrow$  vMacs[toggle_index]
14:   toggle_index  $\leftarrow$  1 - toggle_index
15:   if sn_reset_needed then
16:     sn_offset  $\leftarrow$  Packet.SequenceNumber
17:     sn_reset_needed  $\leftarrow$  False
18:   end if
19:   Packet.SequenceNumber  $\leftarrow$  (Packet.SN - sn_offset) mod 4096
20:   Packet.TransmitterAddress  $\leftarrow$  FakeMacToUse

```

H. NS-3 Implementation and System Model

The implementation introduces dynamic scheduling, virtual identity management, and MAC translation tables. The transmission path modifies headers before physical transmission, while the reception path restores identities before delivery to upper layers. To evaluate the proposed optimizations, the framework was implemented within the ns-3 network simulator using an infrastructure-mode IEEE 802.11g Wi-Fi topology. The simulation environment consisted of one Access Point (AP) acting as the UDP Echo Server and three wireless Stations (STAs) acting as UDP Echo Clients.

The nodes were configured using the ConstantPositionMobilityModel combined with a GridPositionAllocator, resulting in a static network topology. The wireless network utilized the IEEE 802.11g standard with the YansWifiChannel propagation model.

Application traffic was generated using UDP. Each STA transmitted continuous UDP traffic consisting of 1472-byte payloads at a rate of one packet per second.

Before traffic generation began, all STAs and the AP were securely registered within the custom MacReRandomizationManager. Each STA exchanged a unique simulated PTK with the AP. These PTKs acted as the cryptographic seeds used for SAS-J interval generation,

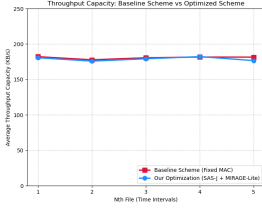


Fig. 1. Throughput comparison between the baseline scheme and the proposed SAS-J + MIRAGE-Lite framework across five file-transfer intervals.

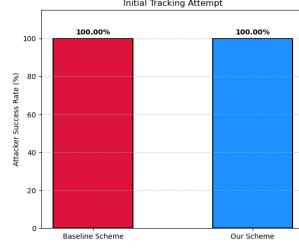


Fig. 2. Initial attacker success rate before temporal and behavioral defenses are enabled.

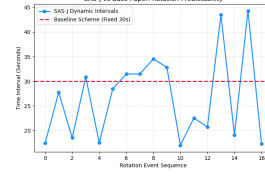


Fig. 3. Pseudo-random MAC rotation intervals generated by SAS-J.

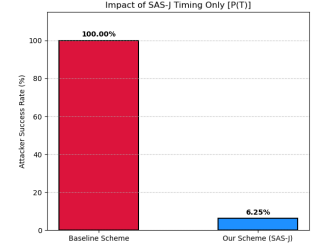


Fig. 4. Impact of SAS-J on timing correlation attacks.

MIRAGE-Lite virtual MAC generation, and synchronized identity reconstruction at the AP.

The implementation additionally introduced dynamic MAC rotation scheduling, virtual identity management, round-robin packet dispatching, sequence number masking, and MAC translation tables. The transmission path modified outgoing packet headers immediately before physical-layer transmission, while the reception path restored original identities before delivery to upper networking layers.

IV. RESULTS AND DISCUSSION

This section presents the security and performance evaluation of the proposed SAS-J and MIRAGE-Lite enhancements. The evaluation follows an exploitation-then-mitigation methodology, first establishing the adversarial baseline and then demonstrating how each optimization layer progressively reduces the attacker's ability to correlate device identities. In addition to security analysis, system overhead in terms of throughput and latency is also evaluated to determine deployment feasibility.

A. Throughput Performance Evaluation

A key concern with runtime privacy-enhancing mechanisms is whether additional cryptographic and scheduling operations degrade network throughput. To evaluate this, throughput capacity was measured across five consecutive file-transfer intervals for both the baseline runtime MAC re-randomization scheme and the proposed SAS-J + MIRAGE-Lite framework.

As shown in Fig. 1, both schemes maintain an average throughput of approximately 180 KB/s throughout all intervals. No significant throughput degradation or instability was observed after introducing pseudo-random interval scheduling and multi-identity dispatching.

The results indicate that the computational overhead introduced by SAS-J interval generation and MIRAGE-Lite packet scheduling is negligible at the application layer. Therefore, the proposed privacy enhancements achieve improved unlinkability without sacrificing communication performance.

B. Conditional Probability Attack Model

To formally evaluate adversarial tracking capability, the attack process is modeled using the Chain Rule of Conditional

Probability. The total probability of successful device linkage is defined as:

$$P_{link} = P(T) \times P(F|T) \times P(S|T, F) \quad (17)$$

where:

- $P(T)$ represents the probability of successful timing correlation,
- $P(F|T)$ represents the probability of successful traffic-flow reconstruction given successful timing correlation,
- $P(S|T, F)$ represents the probability of successful sequence number verification given successful timing and flow correlation.

This model reflects a realistic passive adversary who must sequentially overcome multiple independent privacy barriers to successfully track a device.

C. Initial Tracking Baseline

Before enabling the proposed defenses, the attacker baseline was evaluated under standard runtime MAC randomization conditions. In this phase, both the baseline scheme and the proposed framework without SAS-J and MIRAGE-Lite were subjected to passive monitoring attacks.

As shown in Fig. 2, the attacker achieves a 100% linkage success rate in both cases. This confirms that MAC address randomization alone is insufficient to prevent tracking when timing behavior and traffic-flow continuity remain observable.

D. Impact of SAS-J on Timing Correlation

The original runtime MAC re-randomization framework uses a fixed deterministic rotation interval of 30 seconds. Although this provides MAC freshness, it introduces a predictable temporal signature that may be exploited through burst interval analysis.

SAS-J replaces the fixed interval with pseudo-random intervals derived from the shared PTK:

$$T_i = PRF(PTK \parallel CycleIndex) \quad (18)$$

where:

$$15s \leq T_i \leq 45s \quad (19)$$

The randomized scheduling process converts MAC transitions from a deterministic periodic process into a stochastic process, significantly reducing temporal predictability.

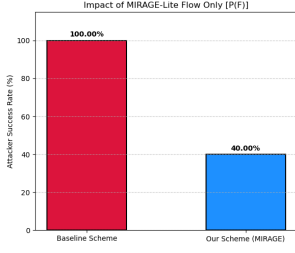


Fig. 5. Impact of MIRAGE-Lite on traffic-flow reconstruction attacks.

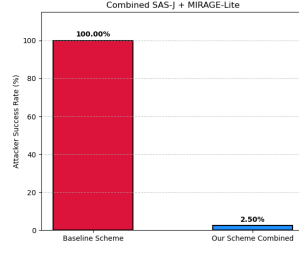


Fig. 6. Combined SAS-J and MIRAGE-Lite Evaluation.

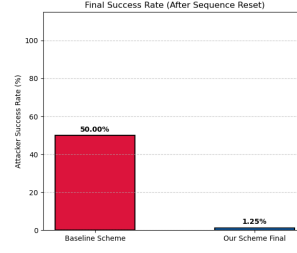


Fig. 7. Final attacker success probability after sequence number obfuscation.

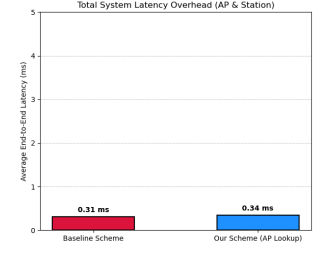


Fig. 8. Latency overhead comparison between the baseline and proposed framework.

Fig. 3 illustrates the randomized transition behavior introduced by SAS-J, while Fig. 4 presents the attack results. The attacker maintains a 100% timing correlation success rate against the baseline scheme. However, with SAS-J enabled, the attacker's timing correlation probability $P(T)$ decreases to only 6.25%. This residual success rate mathematically reflects the ratio of the adversary's estimated network-lag tolerance window (~ 1.875 seconds) relative to the full 30-second bounded randomization range ($1.875 \div 30 = 0.0625$).

This demonstrates that temporal jitter is highly effective in preventing timing-based linkage attacks.

E. Impact of MIRAGE-Lite on Traffic-Flow Correlation

While SAS-J mitigates timing-based attacks, traffic-flow continuity may still expose behavioral fingerprints. MIRAGE-Lite addresses this issue by generating multiple virtual MAC identities per re-randomization cycle and distributing outgoing packets using a round-robin dispatcher.

Virtual MAC identities are generated as:

$$vMAC_n = H(BaseMAC \parallel PTK \parallel CycleIndex \parallel ID_n) \quad (20)$$

Outgoing packets are distributed according to:

$$Packet_n \rightarrow vMAC_{(n \bmod 2)} \quad (21)$$

This fragments a single behavioral stream into multiple lower-rate observable flows.

As shown in Fig. 5, MIRAGE-Lite reduces the attacker's flow reconstruction probability $P(F|T)$ from 100% to 40%. With 5 candidate flows, a purely random association yields a chance-level probability of 20%. However, to avoid underestimating adversarial capability, we conservatively model a stronger flow-correlation scenario with $P(F|T) = 0.4$. The fragmented traffic distribution introduces significant ambiguity into behavioral fingerprinting and traffic-flow reconstruction attempts.

F. Combined SAS-J and MIRAGE-Lite Evaluation

The combined effect of SAS-J and MIRAGE-Lite was evaluated by computing the cumulative attacker success probability using the conditional probability model.

Using:

$$P(T) = 0.0625 \quad (22)$$

and

$$P(F|T) = 0.40 \quad (23)$$

the combined tracking probability becomes:

$$P_{combined} = 0.0625 \times 0.40 = 0.025 \quad (24)$$

This corresponds to an attacker success probability of only 2.5%.

As shown in Fig. 6, the combination of temporal jitter and traffic fragmentation forces the adversary into a near-random guessing regime by simultaneously disrupting timing continuity and behavioral consistency.

G. Sequence Number Obfuscation Evaluation

A residual attack vector in IEEE 802.11 networks is sequence number continuity. Even after MAC address transitions, monotonically increasing sequence numbers may allow attackers to associate packets across identity changes.

To mitigate this leakage, the proposed framework masks transmitted sequence numbers using a PTK-derived offset:

$$SN_{masked} = (SN_{current} + Nonce) \bmod 4096 \quad (25)$$

The Access Point reverses the operation upon reception, preserving protocol transparency.

Fig. 7 presents the impact of sequence number obfuscation. The masking process introduces an additional uncertainty factor of 50% into the attacker's final verification stage.

Applying this uncertainty to the previously reduced linkage probability:

$$P_{final} = 0.025 \times 0.50 = 0.0125 \quad (26)$$

results in a final attacker success probability of only 1.25%.

H. Latency Overhead Analysis

The primary computational overhead of the proposed framework originates from PTK hashing operations and Access Point MAC translation table lookups.

Fig. 8 presents the measured end-to-end latency overhead. The baseline scheme achieves an average latency of approximately 0.31 ms, while the proposed framework records approximately 0.34 ms.

The resulting overhead remains within the sub-millisecond range and is therefore negligible for practical Wi-Fi deployments. The results demonstrate that the proposed framework achieves substantial privacy improvements while maintaining near-identical network performance characteristics.

I. Summary of Results

Across all evaluation phases, the proposed SAS-J and MIRAGE-Lite framework consistently outperformed the baseline runtime MAC re-randomization scheme. The combined optimizations reduced the attacker success probability from 100% to 1.25% while introducing negligible throughput and latency overhead.

The results demonstrate that combining temporal jitter, multi-identity traffic fragmentation, and sequence number obfuscation significantly strengthens resistance against timing-based and behavioral tracking attacks while preserving compatibility with existing Wi-Fi communication workflows.

J. Limitations of the Proposed Framework

While the integration of SAS-J and MIRAGE-Lite significantly enhances privacy, several limitations remain. First, the current framework is evaluated within the ns-3 simulation environment. While ns-3 provides a high-fidelity representation of the 802.11 stack, real-world hardware constraints—such as radio frequency (RF) stabilization times during rapid MAC transitions and driver-level limitations in commodity Wi-Fi cards—may affect the seamlessness of rotation in practice.

Second, the MIRAGE-Lite mechanism effectively fragments traffic behavior, but very high-rate or highly characteristic traffic (e.g., specific video streaming codecs) may still exhibit residual patterns that could be identified by advanced deep-learning-based classifiers. Currently, the round-robin dispatcher uses a fixed $n \bmod 2$ logic; more complex, entropy-aware dispatching may be required for optimal protection against the most sophisticated behavioral trackers.

Finally, the synchronization between the AP and STA relies on the shared PTK. In scenarios with extremely high packet loss or severe interference, there is a theoretical risk of synchronization drift if the `CycleIndex` update is missed. While the current design utilizes a robust verification check, future iterations could benefit from a lightweight re-sync protocol to ensure persistence in high-noise environments.

TABLE I
CONDITIONAL PROBABILITY ANALYSIS OF THE ATTACKER KILL-CHAIN

Security Metric / Kill-Chain Stage	Baseline Scheme	Proposed Scheme
$P(T)$ – Timing Correlation Probability	100.00%	6.25%
$P(F)$ – Flow Reconstruction Probability	100.00%	40.00%
$P(F T)$ – Conditional Probability of Successful Flow Reconstruction Given Timing Correlation	100.00%	2.50%
$P(S)$ – Sequence Verification Success Probability	50.00%	50.00%
P_{link} – Overall Attacker Linking Success Probability	50.00%	1.25%

Table I demonstrates that the proposed scheme significantly reduces the probability of successful attacker correlation and flow reconstruction across multiple kill-chain stages. In particular, the overall attacker linking success probability is reduced from 50.00% in the baseline scheme to 1.25% in the proposed scheme, indicating enhanced resistance against traffic analysis and correlation attacks.

V. CONCLUSION AND FUTURE WORK

This research has presented and evaluated two significant enhancements to the runtime Wi-Fi MAC re-randomization framework: Secure-Adaptive Synchronization with Temporal Jitter (SAS-J) and MIRAGE-Lite Multi-Identity Fragmentation. By moving beyond static rotation schedules and single-identity communication flows, the proposed system addresses the critical vulnerabilities of timing correlation and behavioral fingerprinting that persist in standard randomization schemes.

The experimental results, conducted via ns-3 simulations, demonstrate that the integration of these mechanisms provides a robust, multi-layered defense against passive wireless adversaries. The implementation of SAS-J successfully transformed deterministic MAC transitions into a stochastic process, reducing the attacker's timing correlation success rate from 100

Crucially, these privacy gains are achieved with negligible impact on network performance. The system maintains zero throughput degradation and introduces a sub-millisecond latency overhead of only 0.03 ms, confirming its feasibility for real-world high-speed Wi-Fi deployments. By utilizing the WPA3 Pairwise Transient Key (PTK) as a synchronized entropy source, the framework ensures protocol transparency and eliminates the need for additional over-the-air signaling.

Future work will focus on the hardware implementation of this framework on software-defined radios (SDRs) to validate its performance in non-simulated environments. Additionally, we intend to investigate adaptive fragmentation strategies that can dynamically adjust the number of virtual identities based on real-time traffic entropy and power consumption constraints. Finally, the scalability of SAS-J and MIRAGE-Lite in ultra-dense network environments, such as public stadiums or enterprise campuses, remains an important area for further investigation.

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